



◆ OFFENSIVE SECURITY SERIES ◆

ANDROID SETUP

ADB, Genymotion, Burp CA, Frida & Objection

ADB · Genymotion · Burp CA · Frida Server · Objection · MobSF

ADB Connect

APK Pull

Burp CA Install

Frida Server

Objection

SSL Unpin

Root Bypass

RedTeamGarage (RTG) | Offensive Security Education

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01 — ADB SETUP

Device Connection & APK Pull

A complete Android security lab requires ADB, an emulator or rooted device, Burp Suite with CA cert installed, and Frida/Objection for runtime analysis. This section walks through every setup step with real command output.

```
ADB - Device Connection & APK Pull

# Install ADB and check connected device
(kali)-[~]
kali $ sudo apt install adb && adb devices
List of devices attached
192.168.56.101:5555 device

# List installed packages - find target app
(kali)-[~]
kali $ adb shell pm list packages -3 | grep target
package:com.target.bankingapp
package:com.target.mobileapp

# Pull APK from device
(kali)-[~]
kali $ adb shell pm path com.target.bankingapp
package:/data/app/com.target.bankingapp-XyZ123/base.apk
(kali)-[~]
kali $ adb pull /data/app/com.target.bankingapp-XyZ123/base.apk ./bankingapp.apk
/data/app/com.target.bankingapp-XyZ123/base.apk: 1 file pulled, 0 skipped. 4.2 MB/s (8654321 bytes in 1.967s)

# Verify root access on device
(kali)-[~]
kali $ adb shell su -c id
uid=0(root) gid=0(root) groups=0(root)
```

Figure: ADB connects to Genymotion emulator, target APK pulled from device (4.2 MB)



02 — BURP CA

System Cert Install

```

Burp CA Install as System Cert

# Install Burp CA as system cert (root required)
(kali)-[~]
kali $ openssl x509 -inform DER -in burp_ca.der -out burp_ca.pem
(kali)-[~]
kali $ HASH=$(openssl x509 -inform PEM -subject_hash_old -in burp_ca.pem | head -1) && echo $HASH
9a5ba575
(kali)-[~]
kali $ cp burp_ca.pem 9a5ba575.0
(kali)-[~]
kali $ adb push 9a5ba575.0 /sdcard/
9a5ba575.0: 1 file pushed, 0 skipped. 18.3 MB/s (1456 bytes in 0.000s)
(kali)-[~]
kali $ adb shell su -c "mount -o rw,remount /system && cp /sdcard/9a5ba575.0 /system/etc/security/cacerts/ && chmod 644 /system/etc/security/cacerts/9a5ba575.0"

(kali)-[~]
kali $ adb reboot && sleep 15 && adb shell
* daemon not running; starting now at tcp:5037
* daemon started successfully

# Set proxy on device to point at Burp
(kali)-[~]
kali $ adb shell settings put global http_proxy 192.168.56.1:8080

# Verify – launch app and check Burp HTTP history
(kali)-[~]
kali $ adb logcat | grep -i "http\|proxy\|connect" | head -5
I/NetworkSecurityConfig: No Network Security Config specified, using platform default
D/ConnectivityService: reportNetworkConnectivity(101, true)
```

Figure: Burp CA hash 9a5ba575 installed to /system/etc/security/cacerts/ — HTTPS intercepted



03 — FRIDA SERVER

Push & Start

```
Frida Server – Install & Verify

# Install Frida and push frida-server to device
(kali)-[~]
kali $ pip install frida-tools frida
Successfully installed frida-16.2.1 frida-tools-12.3.0
(kali)-[~]
kali $ adb shell getprop ro.product.cpu.abi
arm64-v8a
(kali)-[~]
kali $ wget https://github.com/frida/frida/releases/download/16.2.1/frida-server-16.2.1-android-arm64.xz -q
(kali)-[~]
kali $ xz -d frida-server-16.2.1-android-arm64.xz && mv frida-server-16.2.1-android-arm64 frida-server
(kali)-[~]
kali $ adb push frida-server /data/local/tmp/ && adb shell chmod 755 /data/local/tmp/frida-server
frida-server: 1 file pushed, 0 skipped. 22.1 MB/s (44892160 bytes in 1.930s)

# Start frida-server on device (root)
(kali)-[~]
kali $ adb shell su -c "/data/local/tmp/frida-server &" && sleep 2

# Verify connection from Kali
(kali)-[~]
kali $ frida-ps -U | grep com.target
12847  com.target.bankingapp
13201  com.target.mobileapp
```

Figure: frida-server v16.2.1 running on arm64 device, target apps visible in frida-ps



04 — OBJECTION

Attach & SSL Unpin

```

Objection – Attach & SSL Unpin

# Install Objection and attach to target app
(kali)-[~]
kali $ pip install objection
Successfully installed objection-1.11.0
(kali)-[~]
kali $ objection -g com.target.bankingapp explore
Using USB device 'Android Emulator 5554'
Agent injected and responds ok!

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Runtime Mobile Exploration
by: @leonjza from @sensepost

[tab] for command suggestions
com.target.bankingapp on (Android: 12) [usb] #

# Disable SSL pinning immediately
(kali)-[~]
kali $ android sslpinning disable
(agent) Custom TrustManager ready
(agent) OkHTTP 3.x CertificatePinner hooked
(agent) Fabric OkHTTP CertificatePinner hooked
(agent) TrustManagerImpl hooked
(agent) SSLContext init() patched with null TrustManager
(agent) Unpinning successful
  
```

Figure: Objection hooks all cert pinning methods — Burp now intercepts all HTTPS traffic



05 — CHEAT SHEET

Quick Reference

Install ADB	<code>sudo apt install adb</code>
Connect emulator	<code>adb connect 192.168.56.101:5555</code>
List 3rd party apps	<code>adb shell pm list packages -3</code>
Find APK path	<code>adb shell pm path com.pkg.name</code>
Pull APK	<code>adb pull /data/app/com.pkg-XXX/base.apk ./app.apk</code>
Root shell	<code>adb shell su -c "id"</code>
Install Burp CA	<code>openssl x509 -inform DER -in burp.der -out burp.pem → HASH=\$(openssl x509 -inform PEM -subject_hash_old -in burp.pem head -1) → cp burp.pem \${HASH}.0 → push to /system/etc/security/cacerts/</code>
Set proxy	<code>adb shell settings put global http_proxy ATTACKER_IP:8080</code>
Clear proxy	<code>adb shell settings put global http_proxy :0</code>
Push frida-server	<code>adb push frida-server /data/local/tmp/ && adb shell chmod 755 /data/local/tmp/frida-server</code>
Start frida-server	<code>adb shell su -c "/data/local/tmp/frida-server &"</code>
Verify frida	<code>frida-ps -U</code>
Attach objection	<code>objection -g com.pkg.name explore</code>
SSL unpin	<code>android sslpinning disable (in objection)</code>
Root bypass	<code>android root disable (in objection)</code>
Logcat secrets	<code>adb logcat grep -iE "password token key bearer"</code>
Screenshot	<code>adb shell screencap /sdcard/screen.png && adb pull /sdcard/screen.png</code>
Install APK	<code>adb install target.apk</code>
MobSF Docker	<code>docker run -it -p 8000:8000 opensecurity/mobile-security-framework-mobsf</code>



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